

PROJECT SYNOPSIS

Title of the Project:

TrackMyFees

Project Definition:

TrackMyFees is a Fee Tracker - a Python and MySQL-based application built to store, manage, and track fee-related records of students enrolled in a coaching institute or a private tutoring setup. The system is designed for two kinds of users: institute staff or private tutors who need to manage their own student records, and parents who pay fees to multiple teachers across different subjects and need a reliable way to keep track of what has been paid, for which month, and to whom. The application handles fee collection, payment history, outstanding dues, and structured summaries through a menu-driven command-line interface. MySQL handles all persistent storage. Python handles all application logic. No external graphical libraries are used.

Contribution / Team Members:

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Reason for Choosing the Topic:

Fee management sounds simple until you are actually in the middle of it. A parent whose child attends four or five different coaching teachers across subjects rarely has a clear picture of which months have been paid to which teacher. A very common situation: March's fee gets paid in March, April slips by unnoticed. When the parent or teacher checks the payment history a few weeks later, March shows up as paid and April does not, and now both sides have to figure out when the last payment was made, how much is still owed, and whether anything was double-counted. Doing this across three or four teachers, for multiple students, is tedious and almost always leads to disputes. At the institute level, the same problems appear in a different form - misplaced records, forgotten dues, and no clear picture of who has paid and who has not. A computerised tracker removes most of this friction. It records every payment against a specific student and date, keeps a clear month-by-month history, and flags outstanding dues automatically, so neither the teacher nor the parent has to manually compare old receipts or rely on memory.

Objective:

- To store and manage student enrollment and fee records in a MySQL database.

- To record fee payments with dates and track outstanding dues per student, per month.
- To let parents check a clear payment history across different teachers and subjects, so they can see at a glance which months are cleared and which are not.
- To generate fee receipts and display individual transaction histories.
- To identify students with pending or overdue payments and flag them.
- To allow adding, updating, and deleting student and fee records as needed.
- To compute total fees collected and total dues pending across all students.
- To support private tutors managing their own small student rosters, not just large institutes.
- To provide a simple, menu-driven text-based interface that staff, tutors, and parents can operate without technical knowledge.

Features of the Project:

- Student Registration – Add new students with details such as name, class, contact number, and course enrolled
- Fee Structure Management – Define and store total fee amount for each course or batch
- Fee Payment Entry – Record payments made by students along with date and amount
- Pending Dues Tracker – Automatically calculate and display remaining balance for each student
- Payment History – View complete transaction history for any student
- Multi-Tutor / Multi-Subject Tracking — Track fee records for the same student across multiple private teachers — useful for parents managing payments to several coaching teachers at once
- Search Student Records – Search by student name, ID, or course to retrieve fee details
- Update and Delete Records – Modify student information or remove records when required
- Fee Collection Summary – Display total fees collected and total dues outstanding for a selected period
- Defaulter List – Generate a list of students who have not paid fees or have dues beyond a set limit
- Data Persistence using MySQL – All records are permanently stored in a MySQL database using Python's mysql-connector module

Hardware Requirements:

A Computer / Laptop with

1. Minimum 4 GB RAM
2. At least 5 GB free disk space
4. Windows 7 or above / Linux / macOS
5. x86 or x64 bit CPU (Intel / AMD architecture)

Software Requirements:

1. Python 3.6 or higher version
2. MySQL Server 8.0 or higher
3. mysql-connector-python library
4. Any Python-supported text editor or IDE (IDLE / VS Code / PyCharm)

Limitations:

- The system is not web-based and uses CLI instead of GUI
- No graphical charts or visual dashboards are used for data representation
- The application does not support online or digital payment integrations
- Multi-user access control and role-based permissions are not implemented
- No automated notification or reminder system for pending fees

References / Bibliography:

- Textbook – Computer Science with Python, Class XII, by Preeti Arora
- Official Python Documentation – <https://docs.python.org>
- MySQL Documentation – <https://dev.mysql.com/doc/>
- Google.com – for examples and reference queries